Arithmetic series

1 For each of the following, find the sum of the first 10 terms of the arithmetic sequence:

a $a = 6$ and $d = 3$

b $a = -18$ and $d = 4$.

c $a = 10$ and $d = 4$

d $a = -4$ and $d = -6$.

2 Find the sum of the first 15 terms of the following arithmetic series:

a $2 + 6 + 10 + \ldots$

b $-17 - 15 - 13 - \ldots$

c $35 + 31 + 27 + \ldots$

3 For each of the following, find the sum of the first 10 terms of the arithmetic sequence:

a $T_1 = 6$ and $T_{10} = 3.5$

b $T_1 = -8$ and $T_{10} = -2.75$

c $T_4 = 4$ and $T_7 = 13$

d $T_6 = 2$ and $T_7 = 5$

4 Find the sum of the first 20 terms of the arithmetic series defined by $T_1 = \pi$ and $T_{20} = 20\pi$.

5 The first term of an arithmetic sequence is -5 and the sixth term is -45.

a Find the value of d.

b Find the sum of the first 14 terms.

6 Find the sum of the first 14 terms in the arithmetic series $13 + \ldots + \dfrac{13}{2}$.

7 Consider the arithmetic series $-9 - 5 - 1 + \ldots + 59$.

a Find n, the number of terms in the series.

b Find the sum of the series.

8 Consider the sum of the odd natural numbers: $1 + 3 + 5 + \ldots$
Show that the sum of the first n odd natural numbers is a perfect square.

9 Find the sum of the first 40 terms of the following arithmetic sequences:

a $T_n = 6n$

b $T_n = 5 - 3n$

10 The first term of an arithmetic sequence is 7, the common difference is 3 and the sum of the

11 Find the sum of all integers between 20 and 50, inclusive, that are divisible by 6.

12 Write the series $1-\frac{1}{2}+\frac{1}{4}-\frac{1}{8}+...+\frac{1}{1024}$ using summation notation.